

Activity-1

Name: Abdulmalik Hussein Al-Haif
Academic No: 445801958
Subject: 462 CCS

S. No:
Section:

EXERCISE NO: 1

TITLE OF EXERCISE: Scikit- Learn

Goal AIM/OBJECTIVE: This activity provides the Scikit Learn fundamentals.

Write the Output

1.

```
# scipy
import scipy
print('scipy: {}'.format(scipy.__version__))

# numpy
import numpy
print('numpy: {}'.format(numpy.__version__))

# matplotlib
import matplotlib
print('matplotlib: {}'.format(matplotlib.__version__))

# pandas
import pandas
print('pandas: {}'.format(pandas.__version__))
```

2. Accessing data using numpy

```
# access values
import numpy
mylist = [[1, 2, 3], [3, 4, 5]]
myarray = numpy.array(mylist)
print(myarray)
print(myarray.shape)
print("First row: %s" % myarray[0])
print("Last row: %s" % myarray[-1])
print("Specific row and col: %s" % myarray[0, 2])
print("Whole col: %s" % myarray[:, 2])
```

3. Draw the Line plot

```
# basic line plot
import matplotlib.pyplot as plt
import numpy
myarray = numpy.array([1, 2, 3])
plt.plot(myarray)
plt.xlabel('some x axis')
plt.ylabel('some y axis')
plt.show()
```

4. Draw Scatter Plot

```
# basic scatter plot
import matplotlib.pyplot as plt
import numpy
x = numpy.array([1, 2, 3])
y = numpy.array([2, 4, 6])
plt.scatter(x,y)
plt.xlabel('some x axis')
plt.ylabel('some y axis')
plt.show()
```

5. DataFrame

```
# dataframe
import numpy
import pandas
myarray = numpy.array([[1, 2, 3], [4, 5, 6]])
rownames = ['a', 'b']
colnames = ['one', 'two', 'three']
mydataframe = pandas.DataFrame(myarray, index=rownames, columns=colnames)
print(mydataframe)
```

6. To load the iris data from a given csv file into a dataframe and print the shape of the data, type of the data and first 3 rows

```
import pandas as pd
data = pd.read_csv("iris.csv")
print("Shape of the data:")
print(data.shape)
print("\nData Type:")
print(type(data))
print("\nFirst 3 rows:")
print(data.head(3))
```

Result:

Library versions:

- scipy: 1.17.1
- numpy: 2.4.6
- matplotlib: 3.11.1
- pandas: 3.0.5

Numpy array:

```
[[1 2 3]
 [3 4 5]]
```

Shape: (2, 3)

First row: [1 2 3]

Last row: [3 4 5]

Specific row and col: 3

Whole column: [3 5]

Iris dataset preview:

sepal.length	sepal.width	petal.length	petal.width	variety
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
4.7	3.2	1.3	0.2	Setosa

Dataset shape: (150, 5)

Dataset dtypes:

sepal.length	float64
sepal.width	float64
petal.length	float64
petal.width	float64
variety	str

